

Optical Electronic Systems – Laboratory Guide

ECE 482/582, Fall 2003

Introduction:

Purpose: The purpose of the lab experiments and the design project is to expose you to the techniques and skills necessary for working with and designing optical electronic systems.

Grading: Your lab write-ups, notebooks, lab quizzes and final design project will determine your grade for the lab. ** Note - You may use your lab notebook as a resource for answering the questions on the quizzes.

Project: You will be required to complete an individual design project by the end of the term. This project will allow you to design your own optical electronic experiment or system and demonstrate it to your peers.

Notebook: Each student will be required to keep a laboratory notebook. The notebook must be BOUND and NUMBERED on each page. You will use the notebook to document your laboratory activities. Notebooks will be randomly checked for completeness throughout the quarter.

Pre-lab Preparation: Each week questions and reading pertaining to the next week's lab will be assigned to prepare you for the required experiment. All required reading and lab preparation material is posted on the course website.

Outline:

Lab 1 – Introduction, safety and measurements
Lab 2 – Photodetectors
Lab 3 – Linewidth Measurements
Lab 4 – Longitudinal Modes
Lab 5 – Gaussian Beams
Lab 6 – Transverse Modes
Lab 7 – Semiconductor Lasers/LEDs
Design Projects (Last 3 weeks)

Location:

The laboratory is located in Dearborn 300. Every student is required to show up to a demonstration/quiz session on Tuesdays. Small groups of students will be assigned 2hr-3hr lab times during the week to complete their weekly lab experiments. **Due to limited lab space you must complete a lab during the week it is assigned (NO MAKEUP LABS).**

Lab Reports

Each student will be required to write-up a short (1-3 page) report answering the assigned questions for the lab experiment and documenting the results of the experiment. The format for these reports will be discussed during the introduction lab session.